Al-Balqa Applied University

جامعة البلقاء التطبيقية

Mid-term Exam

Course Title	General Chemistry lab		Course Code	30206111	
Semester	Summer		Academic Year	2023-2024	
Exam Date	14/8/2024	Day	Wednesday	Time:	2:30 – 3:20
Student Name	[redacted]	Student ID No.		Section	11:30 - 12:30

GÉNERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first10 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:**

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1-8	Multiple Choice	CLIO 1-5	53 %	16		12
1-6	Assay	CLIO 1-5	47 %	14		13
Total			100%	30		25/30
Moderation Review						
Comments						v.good.
Marks Reviewed by					Signature & Date	

Part 1

Question	1	2	3 A	4	5	6 C	7	8
Answer	C	B	C	A	D	B	B	C

Part 2

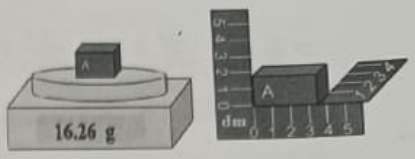
Q	Answer	Q	Answer	Q	Answer
1	Flammable	3	Pycnometer	5	3.416g
2	No affect	4	Solubility	6	Mw = 40g/mol { Calcium Ca

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Q7 – Calculate the density of object (A) :

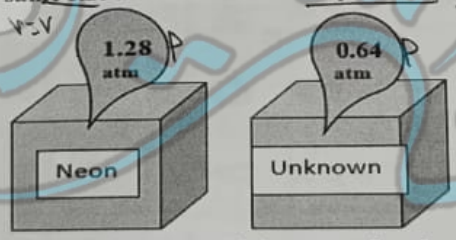
A) 2323 g/cm^3	B) 2.323 x 10^{-3} g/cm^3
C) 2.323 g/cm^3	D) 2.033 g/cm^3
E) 2.033 x 10^{-3} g/cm^3	



Q8- Which one of these salts is an Alum??

A) $NaAl(SO_3)_2.15H_2O$	B) $Na_2B_4O_7 \cdot 10H_2O$	C) $KCr(SO_4)_2.12H_2O$	D) $Be(NO_3)_2 \cdot 4H_2O$	E) A+C

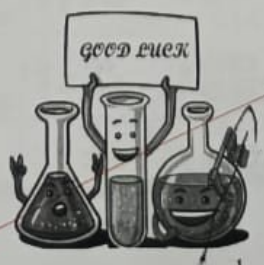
P.2: Answer the following question's and *transfer your answer to the table in the cover page*

Q		Answer
1	**This Pictograms stands for** 	Flammable
2	**If a piece of wood act as impurities in the liquid, this causes the boiling point of liquid to…….**	Not affect
3	**The most precise equipment's to doing the density of liquid experiment is:** (Conical flask , Pycnometer , Graduated cylinder)	Pycnometer
4	**The indication you seen in solubility experiment by mixing a (2mL) of Hexane with some crystals of iodine (I_2) in a test tube is:** non-Solubil + non-Solubil →	Solubility
5	**How many grams of water dose 4.08 g of ($CaCl_2 . 6H_2O$) includes??**	3.4169
6	**The measurements below are for Neon and an unknown gas. The two containers are the same size and at the same temperature.**  **Calculate the molar mass of unknown if the mass of each gas =0.75 g , then name one gas that has a molar mass similar to the one calculated?**	Mm= 40 g/mol Unknown: Calcium Ca

1) Neon
2) Unknown

$T_1 = T_2$
$V_1 = V_2$
$P_1 = 1.28$ atm
$P_2 = 0.64$ atm

$P_1V = n_1RT$
$\left\{\frac{n_1}{P_1} = \frac{V}{RT}\right\}$
$P_2V = n_2RT$
$\left\{\frac{n_2}{P_2} = \frac{V}{RT}\right\} \Rightarrow \frac{n_2}{P_2} = \frac{n_1}{P_1}$



$\frac{m_2}{Mm_2 * P_2} = \frac{m_1}{Mm_1 * P_1}$

$\frac{0.75}{Mm_2 * 0.64} = \frac{0.75}{20 * 1.28}$

$Mm_2 * 0.48 = 19.2$

$Mm_2 = \frac{19.2}{0.48} = 40$ g/mol

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Use the following data when needed:
R=0.08206L.atm/K.mol -- 1atm =760torr =760mmHg =101.3Kpa

P.1 Choose the answer of the following question's and transfer it in CAPITAL letters to table

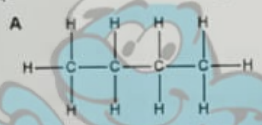
Q1- One of these procedure is incorrect concerning dealing in lab?

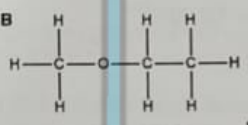
A) Large corrosive spills on clothing may require use of the safety shower.
B) When diluting concentrated acid, add the acid slowly to the water.
C) All coats and backpacks are to be left on the bench-top.
D) Lab coats have to be worn the entire duration of lab time.

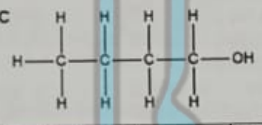
Q2- A solution of silver nitrate ($AgNO_3$) is mixed with a solution of potassium chloride (KCl) . If a precipitate forms, the precipitate is: $AgNO_3 + KCl \longrightarrow KNO_3 + AgCl$

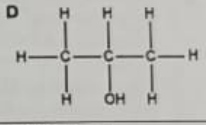
A) KNO_3	B) AgCl	C) No precipitate will form	D) Both products will precipitate

Q3-The correct arrangement of increasing the boiling point of these compounds is:

A 

B 

C 

D 

A) C > D > B > A	B) C > B > D > A	C) C > A > D > B	D) B > C > D > A

Q4- A hydrated salt ($LiCl.xH_2O$) is heated in a crucible until all water has been driven off. If mass percent of water =29.7%. The molar mass (in g/mol) of the hydrated salt equals. ***(At. Masses: Li =7.0, H=1.0, O=16.0 Cl=35.5 g/mol)***

A) 60.5	B) 78.5	C) 270.3	D) 161	E) 216.8

Q5- A student is performing a lab in which he has to figure out the molar mass of diatomic oxygen experimentally. He has a 0.025 g sample of diatomic oxygen in a 17.50 mL beaker and records the temperature of the room to be 27.0 ◦C and the pressure to be 1.000 atm. What molar mass of diatomic oxygen would be calculated using his measurements?

A) 16.00 g/mol	B) 32.00 g/mol	C) 17.58 g/mol	D) 35.17 g/mol	E) 25.32 g/mol

(This question depends on the above ..)

Q6- Another student is performing the same molar mass of diatomic oxygen lab. This student, however, is careless and misreads the thermometer. His measurement for temperature in ◦C is two times that of the first student's. How much will their results differ?

A) Student two's molar mass is exactly two times student one's.	B) Student two's molar mass is lower than the molar mass student one's.
C) Student two's molar mass is higher than the molar mass student one's but not two times.	D) Student one's molar mass is exactly two times student two's